# Diego Méndez Carter

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## Profile

M.Eng. in Robotics, Graphics, and Computer Vision student at UNIZAR focused on perception, planning, and deep learning for autonomous systems. 2.5 years as Sourcing Engineer at Chamberlain Group in electromechanical/electronic component analysis. Experienced in delivering rapid prototypes, conceptual system analysis, and quickly adopting new frameworks across robotics and computer vision projects.

**Skills:** Python · C++ · MATLAB · ROS/ROS2 · PyTorch · Fusion 360 · Eagle · Raspberry Pi · Linux

## Education

**Universidad de Zaragoza (UNIZAR)** *Sep 2025 – Present*
*M. Eng. Robotics, Graphics and Computer Vision*

- **Coursework:** Autonomous Robotics (ROS2), Machine Learning, Computer Vision, SLAM, Deep Learning, Computer Imaging

**National Autonomous University of Mexico (UNAM)** *Aug 2017 – Dec 2022*
*BS in Mechatronics Engineering*

- **Thesis Work:** ***Conceptual Design of a Beach Waste Collection System (2024)*** Here
- **Coursework:** Robotics Foundations, Rapid Prototyping, Product Design, Systems Modeling

## Experience

**Robotics Research Intern** *Universidad de Zaragoza (UNIZAR)* *Zaragoza, Spain*
*Feb 2026 – Jun 2026*

- Testing UR10e arm-cloth interaction in the Genesis physics simulator: developing validation scripts and experiments as a first phase for robotic manipulation research.
- **Hands-on with:** Python, Genesis framework, PyTorch, Linux

**Sourcing Engineer (Electronic Components)** *Chamberlain Group* *Sonora, Mexico*
*Jan 2023 – Jun 2025*

- Lead engineer on shortages and continuity of supply for the sourcing electronics team: circuit/gap analysis, component-level testing across residential/commercial systems (actuators, motors, passives), and cross-functional crisis resolution.
- Delivered 75K USD in productivity projects and resolved 30+ shortage cases.
- Mentored interns through the DEAP program, resolved immediate quality crises achieving long-term changes. Suggested components for new concept designs. Awarded Most Fun Engineer during Engineering Week.

**Digital Art Intern** *Secretariat of Culture – Digital Cultural Center* *Mexico City, Mexico*
*May 2022 – Nov 2022*

- Contributed to the design and testing of a digital art installation for the Cervantino International Festival 2022: chassis concept (Conductos), LED interface testing, and power distribution wiring support.

## Projects

**Master's Selected Implementations**

- **Autonomous Robotics:** Implemented and benchmarked A* and RRT global path planners as Nav2 plugins in ROS2, comparing performance metrics across multiple maps.
- **Computer Vision:** Built a Structure from Motion pipeline to localize a 1950s historical camera within a modern 3D reconstruction using SuperGlue, RANSAC, Essential matrix decomposition, and bundle adjustment.

**CROFI Robotics Club — Projects** [GitHub]

- Built competition robots (line-follower, minisumo, RoboCup Junior soccer) and a beach cleaner prototype using Arduino, Raspberry Pi, OpenCV, ROS, and Fusion 360.